

SF1811 Optimization

Lecture 8: Quadratic functions and unconstrained quadratic optimization

Chapter 9 and selected parts of Chapter 27

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Quadratic functions and unconstrained quadratic optimization

Main question:

How does the matrix H determine the behavior and minimizers of a quadratic function?

Today:

- Quadratic functions and their exact expansion.
- Convexity, existence, and uniqueness of minimizers.
- Positive definite and positive semidefinite matrices.
- Congruence transformations.
- LDL^{T}-factorization.

Quadratic functions

A quadratic function $f : \mathbb{R}^n \to \mathbb{R}$ has the form

$$f(x) = \frac{1}{2} x^{\mathsf{T}} H x + c^{\mathsf{T}} x + c_0,$$

where

$$H = H^{\mathsf{T}} \in \mathbb{R}^{n \times n}, \qquad c \in \mathbb{R}^n, \qquad c_0 \in \mathbb{R}.$$

Following the row-vector convention for the gradient,

$$\nabla f(x) = (Hx + c)^{\mathsf{T}},$$

and the Hessian is

$$F(x) = H.$$

From the preceding lecture:

$$f \text{ is convex on } \mathbb{R}^n \quad \Longleftrightarrow \quad H \succeq 0.$$

For a quadratic function, the Hessian is the constant matrix H.

Why may H be assumed symmetric?

Suppose that a quadratic term is initially written as

$$x^{\mathsf{T}} Q x,$$

where Q is not necessarily symmetric.

Since $x^{\mathsf{T}} Q x$ is a scalar,

$$x^{\mathsf{T}} Q x = (x^{\mathsf{T}} Q x)^{\mathsf{T}} = x^{\mathsf{T}} Q^{\mathsf{T}} x.$$

Therefore,

$$x^{\mathsf{T}} Q x = \frac{1}{2} x^{\mathsf{T}} (Q + Q^{\mathsf{T}}) x.$$

Thus the quadratic form depends only on the symmetric part

$$H = \frac{1}{2}(Q + Q^{\mathsf{T}}).$$

Every quadratic function has a representation with a symmetric matrix.

The exact quadratic expansion

For every $x, d \in \mathbb{R}^n$ and $t \in \mathbb{R}$,

$$\boxed{f(x + td) = f(x) + t(Hx + c)^\mathsf{T} d + \frac{1}{2} t^2 d^\mathsf{T} H d.}$$

The two change terms have different roles:

$$t(Hx + c)^\mathsf{T} d$$

is the first-order change, while

$$\frac{1}{2} t^2 d^\mathsf{T} H d$$

describes the curvature in direction d.

For $t = 1$ and $d = y - x$,

$$f(y) = f(x) + (Hx + c)^\mathsf{T}(y - x) + \frac{1}{2}(y - x)^\mathsf{T} H (y - x).$$

Descent and stationarity

If $(Hx + c)^{\mathsf{T}} d < 0$, then d is a descent direction at x.

Indeed, in

$$f(x + td) - f(x) = t(Hx + c)^{\mathsf{T}} d + \frac{1}{2} t^2 d^{\mathsf{T}} H d,$$

the negative first-order term dominates for positive t sufficiently close to zero.

If

$$Hx + c \neq 0,$$

choose

$$d = -(Hx + c).$$

Then

$$(Hx + c)^{\mathsf{T}} d = -\|Hx + c\|^2 < 0.$$

Hence a necessary condition for $\hat{x}$ to minimize f is $H\hat{x} + c = 0$.

Every minimizer of an unconstrained quadratic function is stationary.

Stationarity is not sufficient

The equation

$$H\hat{x} + c = 0$$

only removes the first-order change.

At a stationary point,

$$f(\hat{x} + td) - f(\hat{x}) = \frac{1}{2}t^2 d^{\mathsf{T}} H d.$$

Thus the type of stationary point depends on the curvature:

- if $d^{\mathsf{T}} H d \geq 0$ for every d, the point is a minimizer;
- if $d^{\mathsf{T}} H d < 0$ for some d, the objective decreases in that direction;
- if $H \succ 0$, the stationary point is the unique minimizer.

Negative curvature

Suppose that $H \not\succeq 0$.

Then there is a direction d such that

$$d^{\mathsf{T}} H d < 0.$$

Along the line $x = td$,

$$f(td) = \frac{1}{2} t^2 d^{\mathsf{T}} H d + t\, c^{\mathsf{T}} d + c_0.$$

The negative quadratic term dominates the linear term as t increases in magnitude. Hence

$$f(td) \longrightarrow -\infty$$

in at least one of the directions $t \to +\infty$ or $t \to -\infty$.

Proposition

If $H \not\succeq 0$, then f is unbounded below and has no minimizer.

The convex case

Suppose that

$$H \succeq 0$$

and that $\hat{x}$ satisfies

$$H\hat{x} + c = 0.$$

For every $x \in \mathbb{R}^n$, the exact quadratic expansion gives

$$\begin{aligned} f(x) - f(\hat{x}) &= (H\hat{x} + c)^{\mathsf{T}}(x - \hat{x}) + \frac{1}{2}(x - \hat{x})^{\mathsf{T}} H (x - \hat{x}) \\ &= \frac{1}{2}(x - \hat{x})^{\mathsf{T}} H (x - \hat{x}) \\ &\geq 0. \end{aligned}$$

Therefore, $\hat{x}$ is a global minimizer.

For a convex quadratic function, stationarity is sufficient for global optimality.

Existence and uniqueness of minimizers

For

$$f(x) = \frac{1}{2}x^{\mathsf{T}}Hx + c^{\mathsf{T}}x + c_0, \qquad H = H^{\mathsf{T}},$$

the following cases may occur:

Condition	Conclusion
$H \not\succeq 0$	f is unbounded below
$H \succeq 0$ and $-c \notin \mathcal{R}(H)$	f is unbounded below
$H \succeq 0$ and $-c \in \mathcal{R}(H)$	one or more minimizers
$H \succ 0$	one unique minimizer

A vector $\hat{x}$ is a minimizer if and only if

$$H \succeq 0 \qquad \text{and} \qquad H\hat{x} = -c.$$

If $H \succ 0$, then $\hat{x} = -H^{-1}c$.

The singular positive-semidefinite case

Suppose that

$$H \succeq 0$$

but

$$-c \notin \mathcal{R}(H).$$

Since H is symmetric,

$$\mathcal{R}(H)^{\perp} = \mathcal{N}(H).$$

Hence there is a nonzero $d \in \mathcal{N}(H)$ such that, after possibly changing its sign,

$$c^{\mathsf{T}} d < 0.$$

Since $Hd = 0$,

$$f(td) = \frac{1}{2} t^2 d^{\mathsf{T}} H d + t\, c^{\mathsf{T}} d + c_0 = t\, c^{\mathsf{T}} d + c_0.$$

Thus

$$f(td) \longrightarrow -\infty \qquad \text{as } t \to +\infty.$$

Positive definite and semidefinite matrices

Let $H = H^{\mathsf{T}} \in \mathbb{R}^{n \times n}$.

The matrix H is positive semidefinite if

$$d^{\mathsf{T}} H d \geq 0 \qquad \text{for every } d \in \mathbb{R}^n.$$

We write $H \succeq 0$.

The matrix H is positive definite if

$$d^{\mathsf{T}} H d > 0 \qquad \text{for every } d \neq 0.$$

We write

$$H \succ 0.$$

For a diagonal matrix $D = \operatorname{diag}(d_1, \ldots, d_n)$,

$$D \succeq 0 \iff d_i \geq 0 \quad \text{for all } i,$$

and

$$D \succ 0 \iff d_i > 0 \quad \text{for all } i.$$

Congruence transformations

Let $B \in \mathbb{R}^{n \times n}$ be nonsingular. Then

$$H \succeq 0 \quad \Longleftrightarrow \quad B^{\mathsf{T}} H B \succeq 0,$$

and

$$H \succ 0 \quad \Longleftrightarrow \quad B^{\mathsf{T}} H B \succ 0.$$

Indeed,

$$v^{\mathsf{T}} B^{\mathsf{T}} H B v = (Bv)^{\mathsf{T}} H (Bv).$$

Since B is nonsingular, Bv runs through all vectors in $\mathbb{R}^n$ as v runs through $\mathbb{R}^n$.

Congruence transformations preserve positive definiteness and semidefiniteness.

LDL^{T}-factorization

For a symmetric positive definite matrix, we seek a factorization

$$H = LDL^{\mathsf{T}},$$

where

- L is unit lower triangular;
- D is diagonal.

Since L is nonsingular,

$$D = L^{-1} H L^{-\mathsf{T}}.$$

Thus D is congruent to H.

Consequently,

$$H \succ 0 \quad \Longleftrightarrow \quad D \succ 0 \quad \Longleftrightarrow \quad d_i > 0, \quad i = 1, \ldots, n.$$

The diagonal entries of D reveal the curvature of H.

One elimination step

Partition the symmetric matrix as

$$H = \begin{pmatrix} h_{11} & h_{21}^{\mathsf{T}} \\ h_{21} & H_{22} \end{pmatrix}, \qquad h_{11} > 0.$$

Define

$$\ell_1 = \frac{h_{21}}{h_{11}}$$

and the Schur complement

$$S = H_{22} - \frac{1}{h_{11}} h_{21} h_{21}^{\mathsf{T}}.$$

Then

$$H = \begin{pmatrix} 1 & 0 \\ \ell_1 & I \end{pmatrix} \begin{pmatrix} h_{11} & 0 \\ 0 & S \end{pmatrix} \begin{pmatrix} 1 & \ell_1^{\mathsf{T}} \\ 0 & I \end{pmatrix}.$$

The same step is then applied recursively to S.

An LDL^{T}-factorization example

Consider

$$H = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

First,

$$d_1 = 2, \qquad \ell_1 = \frac{1}{2}\begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The first Schur complement is

$$S_1 = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} - 2\begin{pmatrix} 1 \\ 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Completing the factorization

For

$$S_1 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

we obtain

$$d_2 = 1, \qquad \ell_2 = 1, \qquad d_3 = 2 - 1^2 = 1.$$

Hence

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

and

$$H = LDL^{\mathsf{T}}.$$

Since all diagonal entries of D are positive,

$$H \succ 0.$$

Solving the optimality equation

For $H \succ 0$, the unique minimizer satisfies

$$H\hat{x} = -c.$$

If

$$H = LDL^{\mathsf{T}},$$

then solve the system in three steps:

1. Forward substitution:
$$Lz = -c.$$
2. Diagonal solve:
$$Dw = z.$$
3. Back substitution:
$$L^{\mathsf{T}}\hat{x} = w.$$

LDL^{T} gives both a definiteness test and an efficient way to compute the minimizer.

Main ideas

After this lecture, you should be able to:

- write and use the exact expansion of a quadratic function;
- identify descent directions from the gradient;
- characterize existence and uniqueness of unconstrained quadratic minimizers;
- explain why negative curvature makes a quadratic function unbounded below;
- define positive definite and positive semidefinite matrices;
- explain why congruence transformations preserve definiteness;
- construct an LDL^{T}-factorization in a small example;
- use the factorization to solve the optimality equation.

The matrix H determines curvature, convexity, existence, uniqueness, and computation.